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Program Resources

LESSON 4-3

Multiplying and Dividing Rational Expressions

Lesson Overview

FOCUS

Objective

Students will be able to:

- ✔ Use the structure of rational expressions to rewrite simple rational expressions in different forms.
- ✔ Understand that rational expressions form a system analogous to the system of rational numbers and use that understanding to multiply and divide rational expressions.

Essential Understanding

Rational expressions form a system similar to the system of rational numbers and can be multiplied and divided by applying the properties of operations as they apply to rational expressions.

COHERENCE

In previous courses, students:

- Multiplied and divided rational numbers.

In this lesson, students:

- Use their understanding of operations with rational numbers to multiply and divide rational expressions.

Increasing Coherence Examples 3–5 are not necessarily expected to be addressed on high stakes assessments.

Later in this topic, students will:

- Rewrite rational expressions to find their sums and differences.

RIGOR

This lesson emphasizes a blend of *procedural skill and fluency* and *application*.

- Students multiply and divide rational expressions using the same procedures they used to multiply and divide rational numbers.
- Students apply what they know about division of rational expressions to solve problems involving surface area and volume.

Glossary is available at SavvasRealize.com.

Vocabulary Builder

REVIEW VOCABULARY — **English** | *Spanish*

- **greatest common factor** | *máximo factor comun*
- **rational number** | *número racional*
- **rational expression** | *expression racional*

NEW VOCABULARY

- **simplified form of a rational expression** | *minima expression de una expression radical*

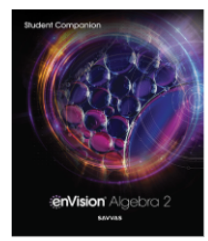
VOCABULARY ACTIVITY

Review the terms *rational number* and *rational expression* with students. Relate the two by discussing how the terms greatest *common factor* and *simplified form* would apply to both rational numbers and rational expressions. Have students match the rational number and rational expression with its simplified form.

$\frac{4}{8}$	$\frac{(x+4)}{(x-3)}$
$\frac{(x+4)(x-2)}{(x-3)(x-2)}$	$\frac{1}{2}$

Student Companion

Students can do their work for the lesson in their *Student Companion* or on Savvas Realize.



Mathematics Overview

Students multiply or divide a rational expression by another rational expression. They also multiply a rational expression by a polynomial by first writing the polynomial as a rational expression with a denominator of 1.

Students write all products and quotients in the simplified form of a rational expression by dividing out all common factors other than 1.

Applying Math Practices

Attend to Precision

Students recognize that a statement of equivalence between two expressions is an identity and that the identity is valid only where both expressions are defined.

Look for and Make Use of Structure

Students recognize that a product of rational expressions can be simplified in the same way as a product of fractions: by dividing the numerator of one fraction and the denominator of the other by their greatest common factor.

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Activity

EXPLORE & REASON

INSTRUCTIONAL FOCUS Students explore a linear function's graph and its domain. They sketch functions that may have restrictions on the domain by creating holes or gaps in their continuity. This prepares students to understand and learn about rational equations, recognizing that the solution can be any real number except those for which the denominator equals 0.

STUDENT COMPANION Students can complete the *Explore & Reason* activity in their *Student Companion*.

Before WHOLE CLASS

Implement Tasks that Promote Reasoning and Problem Solving ETP

Q: What do you notice about the graph of the function? [Sample: *x*-intercept = −2; *y*-intercept = 2; The function is linear.]

During SMALL GROUP

Support Productive Struggle in Learning Mathematics ETP

Q: Why is it important to know the domain of a function? [The domain needs to make sense for the scenario. For example, a real-world problem involving time should remain in Quadrant I on a graph because all numbers should be positive. Also, some values of *x* cause the function to be undefined, so they cannot be included in the domain.]

For Early Finishers

Q: Sketch the function $y = -x + 4$. What is the domain of the function? [all real numbers]

Q: Sketch the graph of the function, excluding −4 from the domain. Describe the new graph created. [The graph is a linear function with a negative slope and an opening at $x = -4$. The *x*-intercept is 4, and the *y*-intercept is 4.]

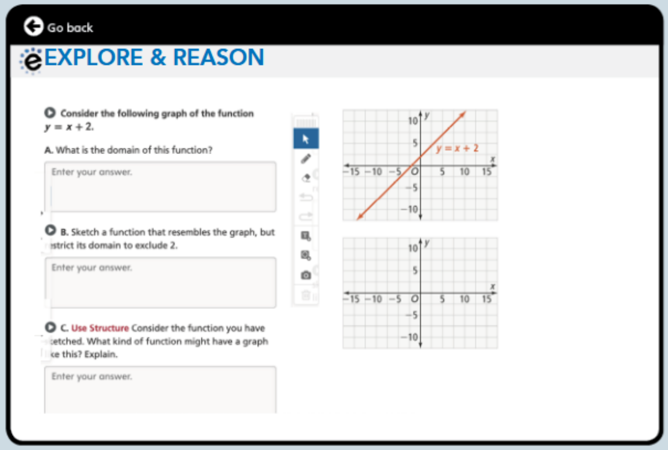
After WHOLE CLASS

Facilitate Meaningful Mathematical Discourse ETP

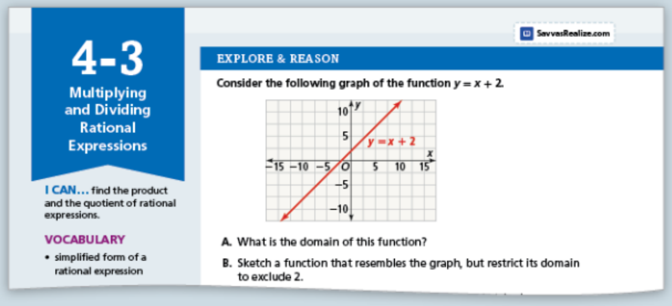
Discuss the difference between functions that have asymptotes and functions that have discontinuities.

Q: How does the exclusion of an *x*-value in a function's domain due to an asymptote compare to the exclusion of an *x*-value in a function's domain at a single point? [An asymptote occurs when there is an expression in a fraction's denominator that equals zero and the numerator is not equal to zero for that value. The graph curves on either side of the asymptote and gets close to but never touches it. When the exclusion is at a single point, the graph appears to have a "hole" at the excluded value, with the function attaining values "close" to the hole.]

Explore & Reason is available at SavvasRealize.com.



STUDENT EDITION


4-3
Multiplying and Dividing Rational Expressions

I CAN… find the product and the quotient of rational expressions.

VOCABULARY
- simplified form of a rational expression

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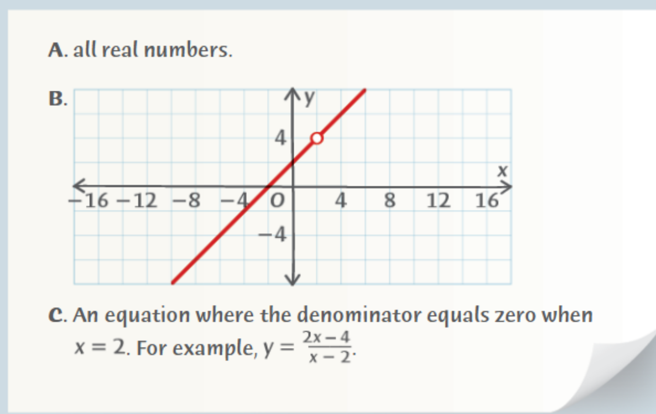
EXPLORE & REASON

Consider the following graph of the function $y = x + 2$.

A. What is the domain of this function?

B. Sketch a function that resembles the graph, but restrict its domain to exclude 2.

SAMPLE STUDENT WORK



Use with **EXPLORE & REASON**

HABITS OF MIND

Reason Does the graph of $y = \frac{2x+6}{x+3}$ have a vertical asymptote at $x = -3$? Explain.

[No; because the function approaches the point (−3, 2) from both sides, there is a hole at $x = -3$ instead of an asymptote.]

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Activity

INTRODUCE THE ESSENTIAL QUESTION

Establish Mathematics Goals to Focus Learning ETP

Introduce students to rational expressions. Explain that multiplying and dividing rational expressions is similar to multiplying and dividing rational numbers.

EXAMPLE 1 Write Equivalent Rational Expressions

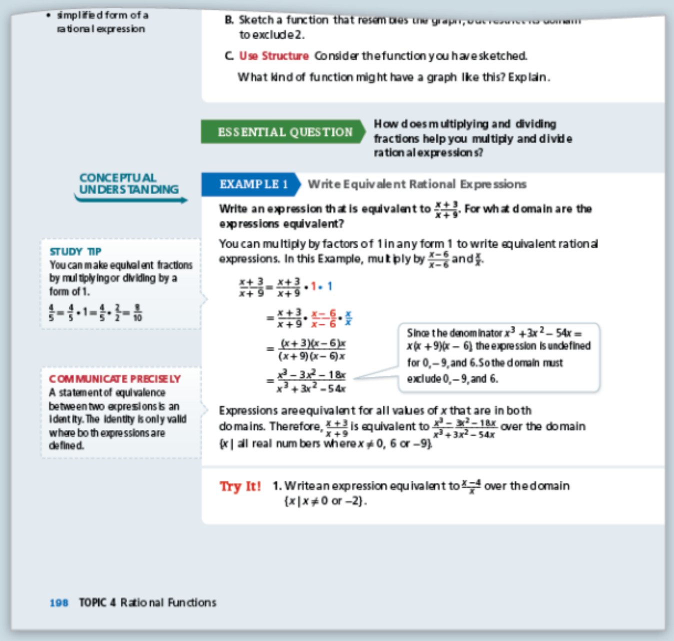
Use and Connect Mathematical Representations ETP

Q: Why is the domain for the identity different from the domain of just the original rational expression? [Because the equivalence of two expressions must occur where the expressions are both defined.]

Q: Why does the denominator affect the domain but the numerator does not?
[For real values of x, a rational expression is the quotient of real numbers. Quotients of real numbers are undefined only when the denominator is 0, and does not depend on the numerator.]

Q: How else can you write an expression equivalent to the original rational expression?
[Multiplying $\frac{x+3}{x+9}$ by any other rational expression, such as $\frac{x}{x}$, $\frac{2-x}{2-x}$, $\frac{x^2}{x^2}$, etc., will result in an equivalent rational expression to the original.]

Examples are available at SavvasRealize.com.



- simplified form of a rational expression

B. Sketch a function that resembles the graph, but restrict its domain to exclude 2.

C. **Use Structure** Consider the function you have sketched. What kind of function might have a graph like this? Explain.

ESSENTIAL QUESTION How does multiplying and dividing fractions help you multiply and divide rational expressions?

CONCEPTUAL UNDERSTANDING

EXAMPLE 1 Write Equivalent Rational Expressions

Write an expression that is equivalent to $\frac{x+3}{x+9}$. For what domain are the expressions equivalent?

You can multiply by factors of 1 in any form 1 to write equivalent rational expressions. In this Example, multiply by $\frac{x-6}{x-6}$ and $\frac{x}{x}$.

$$\frac{x+3}{x+9} = \frac{x+3}{x+9}\cdot 1 \cdot 1$$
$$= \frac{x+3}{x+9}\cdot\frac{x-6}{x-6}\cdot\frac{x}{x}$$
$$= \frac{(x+3)(x-6)x}{(x+9)(x-6)x}$$
$$= \frac{x^3-3x^2-18x}{x^3+3x^2-54x}$$

Since the denominator $x^3+3x^2-54x = x(x+9)(x-6)$, the expression is undefined for 0, −9, and 6. So the domain must exclude 0, −9, and 6.

STUDY TIP
You can make equivalent fractions by multiplying or dividing by a form of 1.
$\frac{4}{5} = \frac{4}{5}\cdot 1 = \frac{4}{5}\cdot\frac{2}{2} = \frac{8}{10}$

COMMUNICATE PRECISELY
A statement of equivalence between two expressions is an identity. The identity is only valid where both expressions are defined.

Expressions are equivalent for all values of x that are in both domains. Therefore, $\frac{x+3}{x+9}$ is equivalent to $\frac{x^3-3x^2-18x}{x^3+3x^2-54x}$ over the domain $\{x \mid$ all real numbers where $x \neq 0, 6$ or $-9\}$.

Try It! 1. Write an expression equivalent to $\frac{x-4}{x}$ over the domain $\{x \mid x \neq 0$ or $-2\}$.

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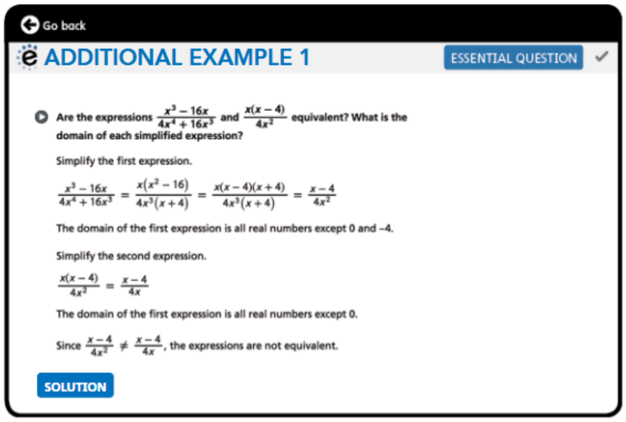
Additional Examples

Additional Examples are available at SavvasRealize.com.



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ADDITIONAL EXAMPLE 1 ESSENTIAL QUESTION

Are the expressions $\frac{x^3-16x}{4x^4+16x^3}$ and $\frac{x(x-4)}{4x^2}$ equivalent? What is the domain of each simplified expression?

Simplify the first expression.

$$\frac{x^3-16x}{4x^4+16x^3} = \frac{x(x^2-16)}{4x^3(x+4)} = \frac{x(x-4)(x+4)}{4x^3(x+4)} = \frac{x-4}{4x^2}$$

The domain of the first expression is all real numbers except 0 and –4.

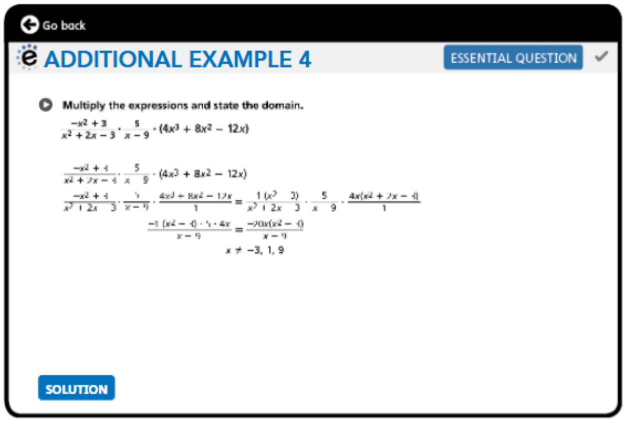
Simplify the second expression.

$$\frac{x(x-4)}{4x^2} = \frac{x-4}{4x}$$

The domain of the first expression is all real numbers except 0.

Since $\frac{x-4}{4x^2} \neq \frac{x-4}{4x}$, the expressions are not equivalent.

SOLUTION



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ADDITIONAL EXAMPLE 4 ESSENTIAL QUESTION

Multiply the expressions and state the domain.

$$\frac{-x^2+3}{x^2+2x-3}\cdot\frac{5}{x-9}\cdot(4x^3+8x^2-12x)$$

$$\frac{-x^2+3}{x^2+2x-3}\cdot\frac{5}{x-9}\cdot(4x^3+8x^2-12x)$$

$$x \neq -3, 1, 9$$

SOLUTION

Example 1 Have students practice identifying mistakes and determining whether rational expressions are equivalent.

Q: Why are 0 and −4 excluded from the domain of the first expression? [These values make the denominator $4x^4 + 16x^3 = 0$, and the denominator cannot equal 0.]

Q: Why is 0 excluded from the domain of the second expression? [This value makes the denominator $4x^3 = 0$, and the denominator cannot equal 0.]

Example 4 Help students transition to multiplying more than one rational expression by a polynomial.

Q: How does multiplying two rational expressions and a polynomial compare with multiplying two rational expressions or a rational expression and a polynomial? [They are all like multiplying fractions. The polynomials are written as rational expressions with a denominator of 1.]

CONTINUED ON THE NEXT PAGE

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Activity

EXAMPLE 1 CONTINUED

Try It! Answer

1. $\dfrac{x^2 - 2x - 8}{x^2 + 2x}$

Elicit and Use Evidence of Student Thinking ETP

Q: What are the common factors that occur in both the numerator and the denominator?
[$x + 2$]

Common Error

Try It! 1 Students may think that a common factor $(x - 2)$ results in excluding -2 from the domain. Have students find the expression or factor when $x = 2$. Then ask them for another factor that could result in excluding -2 from the domain.

EXAMPLE 2 Simplify a Rational Expression

Pose Purposeful Questions ETP

Q: Why is -1 factored out of $2 - x$ in the numerator?
[If you factor out -1, you can divide out the common factor $x - 2$.]

Q: Why can you cross out factors that divide to 1?
[A rational expression with factors that divide to 1 can be written as 1 times another rational expression, and 1 is not typically written in products.]

Try It! Answers

2. a. $\dfrac{x + 1}{x(x - 3)}$ for all x except 0, -1, and 3

b. $x + 1$ for all x except 1 and -4

EXAMPLE 3 Multiply Rational Expressions

Pose Purposeful Questions ETP

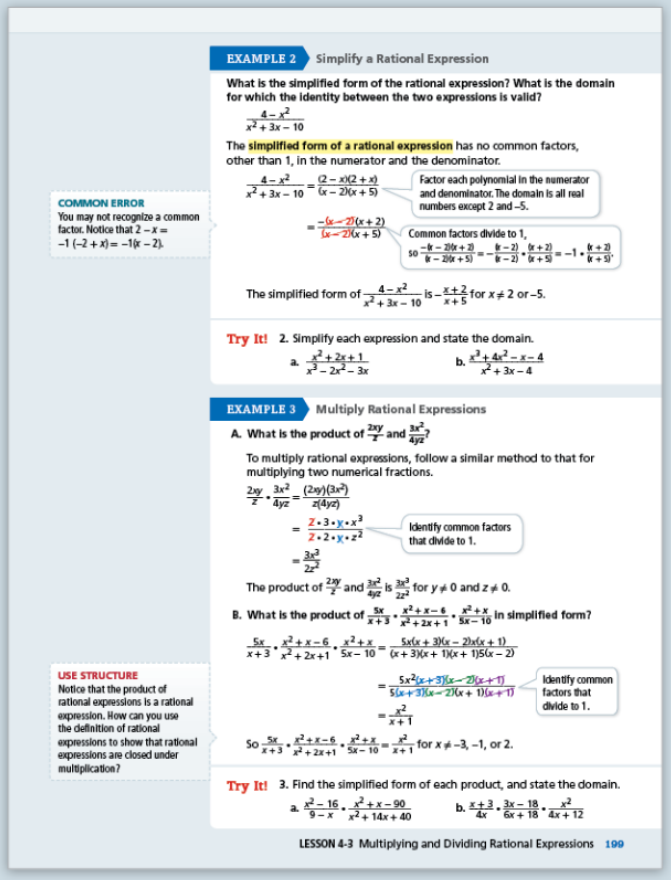
PART A

Q: How can you determine that the domain is $z \neq 0$ and $y \neq 0$ by looking at the rational expressions?
[The denominator cannot equal 0 because the rational expression would be undefined. Therefore, $z \neq 0$ and $y \neq 0$ because multiplying 0 by any number is 0.]

PART B

Q: Why do you factor the expressions in the numerator and the denominator?
[You can divide out the common factors, and you can determine the restrictions on the domain by using the Zero Product Property.]

Examples are available at SavvasRealize.com.



EXAMPLE 2 Simplify a Rational Expression

What is the simplified form of the rational expression? What is the domain for which the identity between the two expressions is valid?

$\dfrac{4 - x^2}{x^2 + 3x - 10}$

The **simplified form of a rational expression** has no common factors, other than 1, in the numerator and the denominator.

$\dfrac{4 - x^2}{x^2 + 3x - 10} = \dfrac{(2 - x)(2 + x)}{(x - 2)(x + 5)}$

Factor each polynomial in the numerator and denominator. The domain is all real numbers except 2 and –5.

$= \dfrac{-\cancel{(x - 2)}(x + 2)}{\cancel{(x - 2)}(x + 5)}$

Common factors divide to 1, so $\dfrac{-(x - 2)(x + 2)}{(x - 2)(x + 5)} = -\dfrac{(x - 2)}{(x - 2)} \cdot \dfrac{(x + 2)}{(x + 5)} = -1 \cdot \dfrac{(x + 2)}{(x + 5)}$.

The simplified form of $\dfrac{4 - x^2}{x^2 + 3x - 10}$ is $-\dfrac{x + 2}{x + 5}$ for $x \neq 2$ or -5.

COMMON ERROR
You may not recognize a common factor. Notice that $2 - x = -1(-2 + x) = -1(x - 2)$.

Try It! 2. Simplify each expression and state the domain.

a. $\dfrac{x^2 + 2x + 1}{x^3 - 2x^2 - 3x}$ b. $\dfrac{x^3 + 4x^2 - x - 4}{x^2 + 3x - 4}$

EXAMPLE 3 Multiply Rational Expressions

A. What is the product of $\frac{2xy}{z}$ and $\frac{3x^2}{4yz}$?

To multiply rational expressions, follow a similar method to that for multiplying two numerical fractions.

$\dfrac{2xy}{z} \cdot \dfrac{3x^2}{4yz} = \dfrac{(2xy)(3x^2)}{z(4yz)}$

$= \dfrac{\cancel{2} \cdot 3 \cdot \cancel{y} \cdot x^3}{\cancel{2} \cdot 2 \cdot \cancel{y} \cdot z^2}$

Identify common factors that divide to 1.

$= \dfrac{3x^3}{2z^2}$

The product of $\frac{2xy}{z}$ and $\frac{3x^2}{4yz}$ is $\frac{3x^3}{2z^2}$ for $y \neq 0$ and $z \neq 0$.

B. What is the product of $\frac{5x}{x+3} \cdot \frac{x^2 + x - 6}{x^2 + 2x + 1} \cdot \frac{x^2 + x}{5x - 10}$ in simplified form?

$\dfrac{5x}{x+3} \cdot \dfrac{x^2 + x - 6}{x^2 + 2x + 1} \cdot \dfrac{x^2 + x}{5x - 10} = \dfrac{5x(x + 3)(x - 2)x(x + 1)}{(x + 3)(x + 1)(x + 1)5(x - 2)}$

$= \dfrac{5x^2\cancel{(x + 3)}\cancel{(x - 2)}\cancel{(x + 1)}}{5\cancel{(x + 3)}\cancel{(x - 2)}(x + 1)\cancel{(x + 1)}}$

Identify common factors that divide to 1.

$= \dfrac{x^2}{x + 1}$

So $\dfrac{5x}{x+3} \cdot \dfrac{x^2 + x - 6}{x^2 + 2x + 1} \cdot \dfrac{x^2 + x}{5x - 10} = \dfrac{x^2}{x + 1}$ for $x \neq -3, -1$, or 2.

USE STRUCTURE
Notice that the product of rational expressions is a rational expression. How can you use the definition of rational expressions to show that rational expressions are closed under multiplication?

Try It! 3. Find the simplified form of each product, and state the domain.

a. $\dfrac{x^2 - 16}{9 - x} \cdot \dfrac{x^2 + x - 90}{x^2 + 14x + 40}$ b. $\dfrac{x + 3}{4x} \cdot \dfrac{3x - 18}{6x + 18} \cdot \dfrac{x^2}{4x + 12}$

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Try It! Answers

3. a. $-(x - 4)$ for all x except 9, -10, and -4

b. $\dfrac{x(x - 6)}{32(x + 3)}$ for all x except 0 and -3

Use with EXAMPLES 1 & 2

HABITS OF MIND

Critique Reasoning Bailey simplified the rational expression $\dfrac{x^2 + 2x + 4}{x^2 + x + 2}$ by dividing out the x^2-terms and then dividing out a factor of $x + 2$ to get 2 as the simplified form of the rational expression. Is Bailey correct? Why or why not?

[No; you cannot eliminate terms from the numerator and denominator if they are bound by addition. You must factor the expression first, and then eliminate like factors.]

Extend Student Thinking

RtI

USE WITH EXAMPLE 3 Challenge students to write an expression as the product of two rational expressions with the given domain.

Q: The domain of the product of two rational expressions is all real numbers except 0, -3, and 6. [Sample: $\dfrac{1}{x^2 + 3x} \cdot \dfrac{1}{x - 6}$]

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Activity

EXAMPLE 4 Multiply a Rational Expression by a Polynomial

Use and Connect Mathematical Representations ETP

Q: What do you already know about multiplying a fraction and a whole number that is helpful here?
[To multiply a fraction and a whole number, you need to also make the whole number a fraction with a denominator of 1.]

Q: Is there a number for x that would make $x^2 + 4 = 0$?
[No; squaring any number, whether negative or positive, results in a positive number. Adding two positive numbers will never equal 0.]

Try It! Answers

4. a. $\dfrac{x^2(x+2)(x-2)(x+1)}{(3x+1)}$ for all x except $\frac{5}{2}$ and $-\frac{1}{3}$

b. $\dfrac{3x(x+2)^2}{(x-7)}$ for all x except 7 and -7

Use with EXAMPLES 3 & 4

HABITS OF MIND

Generalize Why is it important to identify the domain of a rational expression before you simplify it rather than after?

[Simplifying the expression may eliminate some values where the original expression is undefined.]

EXAMPLE 5 Divide Rational Expressions

Pose Purposeful Questions ETP

Q: Why do you factor out -1?
[The factoring out of -1 allows you to divide common factors from the numerator and the denominator.]

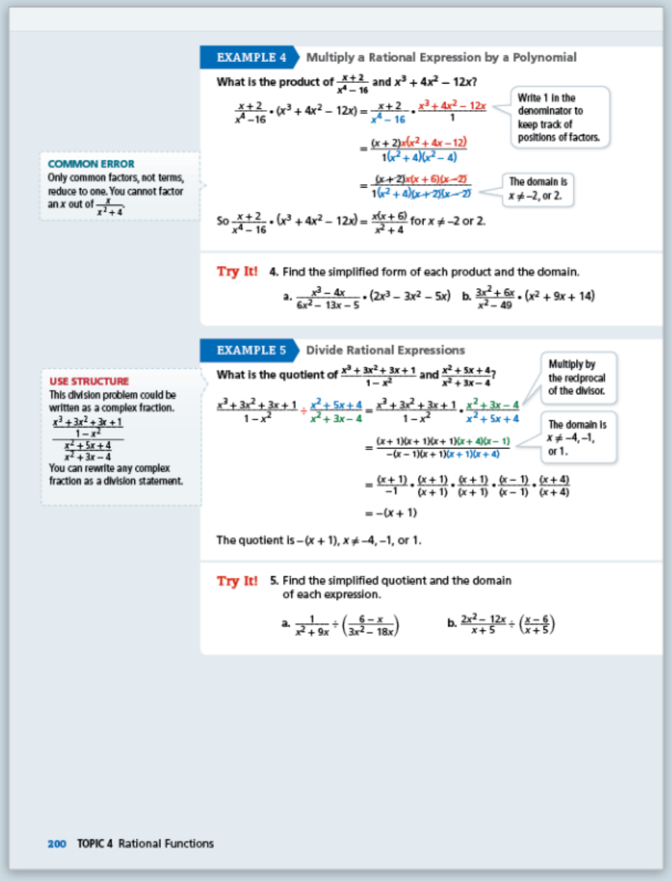
Q: How is dividing rational expressions similar to dividing fractions?
[You multiply by the reciprocal of the divisor.]

Try It! Answers

5. a. $-\dfrac{3}{x+9}$ for all x except 0, -9, and 6

b. $2x$ for all x except 6 and -5

Examples are available at SavvasRealize.com.



EXAMPLE 4 Multiply a Rational Expression by a Polynomial

What is the product of $\frac{x+2}{x^4-16}$ and x^3+4x^2-12x?

$$\frac{x+2}{x^4-16}\cdot(x^3+4x^2-12x) = \frac{x+2}{x^4-16}\cdot\frac{x^3+4x^2-12x}{1}$$

Write 1 in the denominator to keep track of positions of factors.

$$= \frac{(x+2)x(x^2+4x-12)}{1(x^2+4)(x^2-4)}$$

$$= \frac{\cancel{(x+2)}x(x+6)\cancel{(x-2)}}{1(x^2+4)\cancel{(x+2)}\cancel{(x-2)}}$$

The domain is $x \neq -2$, or 2.

So $\frac{x+2}{x^4-16}\cdot(x^3+4x^2-12x) = \frac{x(x+6)}{x^2+4}$ for $x \neq -2$ or 2.

COMMON ERROR
Only common factors, not terms, reduce to one. You cannot factor an x out of $\frac{x}{x^2+4}$.

Try It! 4. Find the simplified form of each product and the domain.

a. $\frac{x^3-4x}{6x^2-13x-5}\cdot(2x^3-3x^2-5x)$ b. $\frac{3x^2+6x}{x^2-49}\cdot(x^2+9x+14)$

EXAMPLE 5 Divide Rational Expressions

What is the quotient of $\frac{x^3+3x^2+3x+1}{1-x^2}$ and $\frac{x^2+5x+4}{x^2+3x-4}$?

$$\frac{x^3+3x^2+3x+1}{1-x^2} \div \frac{x^2+5x+4}{x^2+3x-4} = \frac{x^3+3x^2+3x+1}{1-x^2}\cdot\frac{x^2+3x-4}{x^2+5x+4}$$

Multiply by the reciprocal of the divisor.

$$= \frac{(x+1)(x+1)(x+1)(x+4)(x-1)}{-(x-1)(x+1)(x+1)(x+4)}$$

The domain is $x \neq -4, -1$, or 1.

$$= \frac{(x+1)}{-1}\cdot\frac{(x+1)}{(x+1)}\cdot\frac{(x+1)}{(x+1)}\cdot\frac{(x-1)}{(x-1)}\cdot\frac{(x+4)}{(x+4)}$$

$$= -(x+1)$$

The quotient is $-(x+1)$, $x \neq -4, -1$, or 1.

USE STRUCTURE
This division problem could be written as a complex fraction.
$$\frac{\frac{x^3+3x^2+3x+1}{1-x^2}}{\frac{x^2+5x+4}{x^2+3x-4}}$$
You can rewrite any complex fraction as a division statement.

Try It! 5. Find the simplified quotient and the domain of each expression.

a. $\frac{1}{x^2+9x}\div\left(\frac{6-x}{3x^2-18x}\right)$ b. $\frac{2x^2-12x}{x+5}\div\left(\frac{x-6}{x+5}\right)$

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Support Struggling Students

USE WITH EXAMPLE 5 Students have the opportunity for guided practice in dividing two rational expressions.

Divide $\dfrac{3x^3+6x^2+3x}{x+2} \div \dfrac{x^2+x}{x+2}$.

Multiply by the reciprocal of the divisor. Factor the numerators and denominators individually. Divide out the common factors.

Q: How do you decide what can be factored out of each expression? [Find the greatest common factor.]

Q: What are the quotient and the domain? [$3(x+1)$; $x \neq -2, -1, 0$]

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Activity

EXAMPLE 6 Use Division of Rational Expressions

Examples are available at SavvasRealize.com.

Pose Purposeful Questions ETP

Q: What does it mean to have *the lesser ratio of surface area to volume*?
[the division of surface area by volume that results in a smaller rational expression]

Q: How do the efficiency ratios of the rectangular prism and the cylinder compare to each other?
[The rectangular prisms and the cylinder will always have the same efficiency ratio, regardless of the x-value selected.]

Try It! Answer

6. The efficiency ratio for the prism with the rectangular base is less than that for the prism with the square base.

Use with EXAMPLES 5 & 6

HABITS OF MIND

Use Structure Is the domain of the quotient $\frac{2x^2-12x}{x+5} \div \left(\frac{x-6}{x+5}\right)$ different from the domain of the product $\left(\frac{2x^2-12x}{x+5}\right)\left(\frac{x-6}{x+5}\right)$? Explain.

[Yes; the domain of the product includes the value $x = 6$, while the domain of the quotient does not include $x = 6$.]

APPLICATION

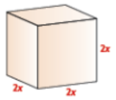
EXAMPLE 6 Use Division of Rational Expressions

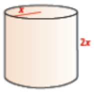
A company is evaluating two packaging options for its product line. The more efficient design will have the lesser ratio of surface area to volume. Should the company use packages that are cylinders or rectangular prisms?

Option 1: A rectangular prism with a square base

Option 2: A cylinder with the same height as the prism, and diameter equal to the side length of the prism's base

STUDY TIP
Recall that surface area tells how much packaging material is needed and volume tells how much product the package can hold.





Surface Area: $2(2x)^2 + 4(2x)^2$
Volume: $(2x)^3$

Surface Area: $2\pi x^2 + 2\pi x(2x)$
Volume: $\pi x^2(2x)$

The efficiency ratio is $\frac{SA}{V}$, where SA represents surface area and V represents volume.

Option 1:

$$\frac{SA}{V} = \frac{2(4x^2) + 4(4x^2)}{8x^3} = \frac{24x^2}{8x^3} = \frac{3}{x}$$

Option 2:

$$\frac{SA}{V} = \frac{2\pi x^2 + 4\pi x^2}{2\pi x^3} = \frac{6\pi x^2}{2\pi x^3} = \frac{3}{x}$$

The company can now compare the efficiency ratio of the package designs.

Prism: $\frac{3}{x}$ Cylinder: $\frac{3}{x}$

In this example, the efficiency ratio of the cylinder is equal to that of the prism. So the company should choose their package design based on other criteria.

Regardless of what positive value is selected for x, the efficiency ratios for these two package designs will be the same.

Try It! 6. The company compares the ratios of surface area to volume for two more containers. One is a rectangular prism with a square base. The other is a rectangular prism with a rectangular base. One side of the base is equal to the side length of the first container, and the other side is twice as long. The surface area of this second container is $4x^2 + 6xh$. The heights of the two containers are equal. Which has the smaller surface area-to-volume ratio?

LESSON 4-3 Multiplying and Dividing Rational Expressions 201

ELL English Language Learners (*Use with* EXAMPLE 6)

SPEAKING BEGINNING When you *evaluate* something, you assign it a value or level of importance. Display a pen, a stapler, a paper clip, and a cell phone. Have students discuss the following questions in small groups.

Q: How you would arrange these items from lowest to highest in value?
[paper clip, pen, stapler, cell phone]

Q: Why would the packaging with a lesser ratio of surface area to volume be more valuable to a company?
[If the company spends less on packaging, they can make a larger profit.]

READING DEVELOPING Have students read the problem statement in the example. Have students point to the words *ratio*, *surface area*, and *volume* as they read them.

Q: What does *surface area* mean?
[the sum of the areas of all of the outside surfaces of an object]

Q: What does *volume* mean?
[the capacity an object can hold, or the amount of space it takes up]

Q: Why is the design with the lesser *ratio* of surface area to volume more efficient?
[because the packaging would use less material to contain a given volume]

LISTENING EXPANDING Distribute a piece of construction paper and a pair of scissors to pairs of students. Ask students to cut the paper into 5 or 6 polygons and mix them up. Have one person put the puzzle pieces back together while the other records the time it takes. Then switch roles. Have students repeat this exercise with their eyes closed. Listen to the definition of *efficient*: "performing in the best possible manner with the least waste of time and effort."

Q: Which method of putting the puzzle together was more *efficient*? Explain
[The first method was more efficient.]

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Presentation
Practice

CONCEPT SUMMARY Products and Quotients of Rational Expressions

Q: How do multiplying or dividing rational expressions compare to multiplying or dividing fractions?
[When multiplying rational expressions and multiplying fractions, you write a whole number and the expression with a denominator of 1 and then identify common factors and simplify. When dividing rational expressions and dividing fractions, you multiply by the reciprocal of the divisor.]

Do You UNDERSTAND? | Do You KNOW HOW?

Common Error

Exercise 9 Students may struggle with division when the divisor is a polynomial instead of a rational expression. Ask students how the polynomial can be written as a rational expression. Then ask students how they can write the quotient of rational expressions as a product of rational expressions.

Answers

1. The steps and operations used to multiply and divide rational expressions are similar to those used with fractions.
2. A rational expression is the ratio of two polynomials that has a domain of all real numbers except those for which the denominator equals zero; Sample: $\frac{3x^2 + 6x}{9x}$
3. The student was supposed to multiply by the reciprocal. The student also did not place any restrictions on the domain. The rational expression should be $\frac{y}{x}$, $x \neq 0$, $y \neq 0$.
4. When the denominator equals 0, fractions are undefined.
5. $\frac{x-6}{x-3}$ for all x except -6 and 3
6. $\frac{xz}{y}$, $x \neq 0$, $y \neq 0$, and $z \neq 0$
7. $\frac{y+2}{y-3}$, $y \neq -3$, 3, or -2
8. $(x-2)(x-4)$ for all x except 5 and -5
9. $\frac{x}{x-4}$ for all x except 4 and -7
10. $\frac{2x(x+4)}{(x+1)}$ for all x except -6, 6, -1, 0, and $\frac{1}{2}$

Concept Summary, Do You Understand, and **Do You Know How** are available at SavvasRealize.com.

CONCEPT SUMMARY Products and Quotients of Rational Expressions

	Multiply	Multiply an Integer or a Polynomial	Divide
RATIONAL EXPRESSIONS	$\frac{3x}{x+1} \cdot \frac{x^2+x}{3x-6}$ The domain is $x \neq -1$ or 2.	$\frac{x+2}{x^2-4} \cdot (x^2 - 2x)$ $= \frac{x+2}{x^2-4} \cdot \frac{x^2-2x}{1}$ The domain is $x \neq -2$ or 2.	$\frac{1-x^2}{x^2+3x-4} \div \frac{x+1}{x+4}$ $= \frac{1-x^2}{x^2+3x-4} \cdot \frac{x+4}{x+1}$ The domain is $x \neq -4$, -1, or 1.
WORDS	Identify common factors and simplify.	Write the polynomial as a rational expression with 1 in the denominator. Then multiply.	Multiply by the reciprocal of the divisor.

Do You UNDERSTAND?

1. ESSENTIAL QUESTION How does multiplying and dividing fractions help you multiply and divide rational expressions?
2. **Vocabulary** In your own words, define *rational expression* and provide an example of a rational expression.
3. **Error Analysis** A student divided the rational expressions as follows:

$$\frac{4x}{5y} \div \frac{20x^2}{25y^2} = \frac{4x}{5y} \div \frac{5 \cdot 4x^2}{25y^2}$$
$$= \frac{16x^3}{25y^3}$$

Describe and correct the errors the student made.

4. **Communicate Precisely** Why do you have to state the domain when simplifying rational expressions?

Do You KNOW HOW?

5. What is the simplified form of the rational expression $\frac{x^2-36}{x^2+3x-18}$? What is the domain?

Find the simplified form of each product, and state the domain.

6. $\frac{4x^2y^2}{3z} \cdot \frac{15z^2}{20xy^3}$
7. $\frac{y+3}{y+2} \cdot \frac{y^2+4y+4}{y^2-9}$
8. $\frac{x^2+3x-10}{x^2-25} \cdot (x^2 - 9x + 20)$

Find the simplified form of each quotient, and state the domain.

9. $\frac{x^3+14x^2+49x}{x^2+3x-28} \div (x+7)$
10. $\frac{6x^4+21x^3-12x^2}{x^3+x^2-36x-36} \div \frac{12x^2-6x}{4x^2-144}$

202 TOPIC 4 Rational Functions

PRACTICE & PROBLEM SOLVING

Practice & Problem Solving and **video tutorials** are available at SavvasRealize.com.

Lesson Practice You may opt to have students complete the automatically scored Practice and Problem Solving items online powered by MathXL for School.

Choose from: **Lesson Practice**
Adaptive Practice

You may also take advantage of the bank of exercises for assigning additional practice.

Assignment Guide

On-Level	Advanced
11–34, 37–40	11–20, 23–40

Engage Through Student Choice

Promote student agency by allowing students to choose practice items. You may structure this choice in many ways.

For example:

Assign each section a point value. Students choose at least one item from each section and items chosen should have a minimum of 20 total points.

Understand, Apply...................2 points each
Practice....................................1 point each
Assessment Practice................1 point each
Performance Task.....................3 points each

Item Analysis

Example	Items	DOK
1	20, 21, 38	1
	11, 14–16, 19	2
2	22–25, 39	1
	17, 38	2
3	26, 27	1
	12, 18	2
4	28, 29	1
5	30–33	1
	40	2
	13	3
6	34, 35	1
	36, 37	2

Scan the QR code to access a video tutorial.

PRACTICE & PROBLEM SOLVING

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UNDERSTAND

11. **Reason** Explain why $\frac{4x^2-7}{4x^2-7}=1$ is a valid identity under the domain of all real numbers except $\pm\frac{\sqrt{7}}{2}$.

12. **Error Analysis** Describe the error a student made in multiplying and simplifying $\frac{x+2}{x-2}\cdot\frac{x^2-4}{x^2+x-2}$.

$$\frac{x+2}{x-2}\cdot\frac{x^2-4}{x^2+x-2}$$
$$=\frac{x+2}{\cancel{x-2}}\cdot\frac{\cancel{(x+2)}\cancel{(x-2)}}{\cancel{(x+2)}(x-1)}$$
$$=\frac{2}{-1}$$

13. **Higher Order Thinking** Why are rational expressions closed under multiplication and division?

14. **Use Appropriate Tools** Explain why domain restrictions are necessary to show that the rational expressions $\frac{-6x^2+21x}{3x}$ and $-2x+7$ are equivalent. What is true about the graphs at $x=0$ and why?

15. **Generalize** Explain the similarities between rational numbers and rational expressions.

16. **Use Structure** Determine whether $\frac{5x+11}{6x+11}=\frac{5}{6}$ is *sometimes*, *always*, or *never* true. Justify your reasoning.

17. **Construct Arguments** Explain how you can tell whether a rational expression is in simplest form.

18. **Communicate Precisely** When multiplying $\frac{15}{x}\cdot\frac{x}{3}=5$, is it necessary to make the restriction $x\neq 0$? Why or why not?

19. **Reason** If the denominator of a rational expression is x^3+3x^2-10x, what value(s) must be restricted from the domain for x?
$-5, 0, 2$

PRACTICE

Write an equivalent expression over the given domain. SEE EXAMPLE 1

20. $\frac{x(x-2)}{(x-5)}$ for all x except 5 and -6

21. $\frac{3x}{x-2}$ for all x except 2 and -5

What is the simplified form of each rational expression? What is the domain? SEE EXAMPLE 2

22. $\frac{y^2-5y-24}{y^2+3y}$ $\frac{(y-8)}{y}$ for all y except 0 and -3

23. $\frac{ab^3-9ab}{12ab^2+12ab-144a}$ $\frac{b(b+3)}{12(b+4)}$ for all real numbers except $a=0$, $b=-4$ and 3

24. $\frac{x^2+8x+15}{x^2-x-12}$ $\frac{x+5}{x-4}$, $x\neq-3, 4$

25. $\frac{x^3+9x^2-10x}{x^3-9x^2-10x}$ $\frac{(x+10)(x-1)}{(x-10)(x+1)}$, $x\neq-1, 0, 10$

Find the product and the domain. SEE EXAMPLE 3

26. $\frac{x^2+6x+8}{x^2+4x+3}\cdot\frac{x+3}{x+2}$

27. $\frac{(x-y)^2}{x+y}\cdot\frac{3x+3y}{x^2-y^2}$

Find the product and the domain. SEE EXAMPLE 4

28. $\frac{(x+5)}{(x^3-25x)}\cdot(2x^3-11x^2+5x)$ $2x-1$ for all x except 0, -5, and 5

29. $\frac{(2x^2-10x)}{(x-5)(x^2-1)}\cdot(3x^2+4x+1)$ $\frac{2x(3x+1)}{x-1}$ for all x except -1, 1, and 5

Find the quotient and the domain. SEE EXAMPLE 5

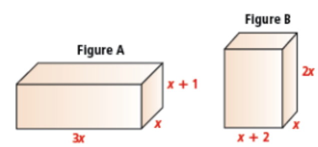
30. $\frac{y^2-16}{y^2-10y+25}\div\frac{3y-12}{y^2-3y-10}$ $\frac{(y+4)(y+2)}{3(y-5)}$ for all y except 4 and 5

31. $\frac{(x-y)^2}{x+y}\div\frac{3x+3y}{x^2-y^2}$

32. $\frac{25x^2-4}{x^2-9}\div\frac{5x-2}{x+3}$ $\frac{5x+2}{x-3}$ for all x except -3, $\frac{2}{5}$, and 3

33. $\frac{x^4+x^3-30x^2}{x^2-3x-18}\div\frac{x^3+x^2-30x}{x^2-36}$ $\frac{x(x+6)}{(x+3)}$ $x\neq-6, -3, 0, 5, 6$

34. Find and simplify the ratio of the volume of Figure A to the volume of Figure B. SEE EXAMPLE 6



$\frac{3(x+1)}{2(x+2)}$ for all $x>0$

Answers

11. Using the difference of two squares formula, $\frac{4x^2-7}{4x^2-7}=\frac{(2x+\sqrt{7})(2x-\sqrt{7})}{(2x+\sqrt{7})(2x-\sqrt{7})}$. Both factors on the numerator and denominator divide to 1, so $\frac{4x^2-7}{4x^2-7}=1$ over all elements of the domain, which is all real numbers except $\pm\frac{\sqrt{7}}{2}$.

12. The student cannot cancel the x's in $x+2$ and $x-1$ because only common factors, not terms, can be canceled.

13. Sample: $\frac{a}{b}\div\frac{c}{d}=\frac{\frac{a}{b}}{\frac{c}{d}}=\frac{\frac{a}{b}}{\frac{c}{d}}\times\frac{\frac{d}{c}}{\frac{d}{c}}=\frac{\frac{ad}{bc}}{\frac{cd}{dc}}=\frac{\frac{ad}{bc}}{\frac{cd}{cd}}=\frac{\frac{ad}{bc}}{1}=\frac{ad}{bc}=\frac{a}{b}\times\frac{d}{c}$

See answers for Exercises 14–18, 20–21, 26–27 and 31 on next page.

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Practice

Answers

14. Sample answer: Domain restrictions are needed because you may not be able to tell just from graphs when two expressions are not equivalent. The graphs of $y = -\frac{6x^2 + 21x}{3x}$ and $y = -2x + 7$ may look identical, but using the trace feature or a table values, $y = -\frac{6x^2 + 21x}{3x}$ is undefined at $x = 0$, while $y = -2x + 7$ is equal to 7 at $x = 0$.

15. A rational number is the ratio of two integers, while a rational expression is a ratio of two polynomials.

16. Never; when you cross multiply, you get $30x + 66 = 30x + 55$; since $66 \neq 55$, there is no real number x for which that equation is true.

17. A rational expression is in simplest form when the numerator and denominator have no common factors other than 1.

18. Yes; even though the factor of x was canceled out, the original fractions cannot be undefined, so the restriction is necessary.

20. $\frac{x^3 + 4x^2 - 12x}{x^2 + x - 30}$

21. $\frac{3x^2 + 15x}{x^2 + 3x - 10}$

26. $\frac{x+4}{x+1}$ for all x except -3, -2, and -1

27. $\frac{3(x-y)}{x+y}$ for all real numbers except $x = y$ or $x = -y$

31. $\frac{(x-y)^3}{3(x+y)}$ for all x except $x = -y$

35. **a.** $\frac{r}{3}$

b. $\frac{3r}{8}$

c. cylinder B; $\frac{3}{8}r > \frac{1}{3}r$

36. The probability is $\frac{x}{6(x+1)}$.

37. The length of the base of the parallelogram is $\frac{3}{5}$ units.

40. **PART A** 9%

PART B 4.6%

Practice & Problem Solving is available at SavvasRealize.com.

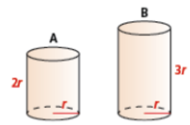
PRACTICE & PROBLEM SOLVING

APPLY

35. Make Sense and Persevere An engineering firm wants to construct a cylindrical structure that will maximize the volume for a given surface area. Compare the ratios of the volume to surface area of each of the cylindrical structures shown, using the following formulas for volume and surface area of cylinders.

Volume $(V) = \pi r^2 h$

Surface Area $(SA) = 2\pi rh + 2\pi r^2$

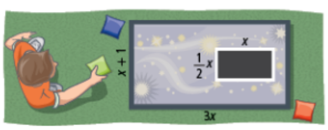


a. Calculate the ratio of volume to surface area for cylinder A.

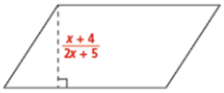
b. Calculate the ratio of volume to surface area for cylinder B.

c. Which of these cylinders has a greater ratio of volume to surface area?

36. Model With Mathematics Lupe designed a carnival game that involves tossing a beanbag into the box shown. In order to win a prize, the beanbag must fall inside the black rectangle. The probability of winning is equal to the ratio of the area of the black rectangle to the total area of the face of the box shown. Find the ratio that represents this probability in simplified form.



37. Look for Relationships A parallelogram with an area of $\frac{3x+12}{10x+25}$ square units has a height shown. Find the length of the base of the parallelogram.



ASSESSMENT PRACTICE

38. Which of the following rational expressions simplify to $\frac{y}{y+3}$? Select all that apply.

□ $\frac{(2y^2 + y)(y+3)}{(4y+2)(y+3)^2}$

■ $\frac{3y^2 + y}{3y^2 + 10y + 3}$

■ $\frac{2y^3 + 3y^2 + y}{(2y+1)(y^2 + 4y + 3)}$

□ $\frac{y^2 + 2y}{y^2 + 4y + 3}$

□ $\frac{\frac{1}{y}}{y+3}$

39. SAT/ACT For what value of x is $\frac{2x^2 + 8x}{(x+4)(x^2-9)}$ undefined?

Ⓐ -8

● -3

Ⓒ 0

Ⓓ 4

Ⓔ 9

40. Performance Task The approximate annual interest rate r of a monthly installment loan is given by the formula:

$$r = \frac{\left[\frac{24(nm - p)}{n}\right]}{\left(p + \frac{nm}{12}\right)},$$

where n is the total number of payments, m is the monthly payment, and p is the amount financed.

Part A Find the approximate annual interest rate (to the nearest percent) for a four-year signature loan of \$20,000 that has monthly payments of \$500.

Part B Find the approximate annual interest rate (to the nearest tenth percent) for a five-year auto loan of \$40,000 that has monthly payments of \$750.

See answers for Exercises 28–30 on next page.

LESSON QUIZ

Use the Lesson Quiz to assess students' understanding of the mathematics in the lesson.
Students can take the Lesson Quiz online or you can download a printable copy from **SavvasRealize.com**. The Lesson Quiz is also available in the *Assessment Sourcebook*.

Item Analysis

Item	DOK	Skills Review and Practice
1	2	A85
2	2	A85
3	2	A85
4	2	A85
5	2	A85

RtI Use the student scores on the Lesson Quiz to prescribe differentiated assignments.

If students take the Lesson Quiz online, it will be automatically scored and appropriate differentiated practice will be assigned based on student performance.

I Intervention	0–3 points	• Reteach to Build Understanding • Mathematical Literacy and Vocabulary • Lesson Virtual Nerd videos
O On-Level	4 points	• Enrichment
A Advanced	5 points	• Enrichment

Lesson Quiz is available at SavvasRealize.com.

Assessment Resources

Name ______

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4-3 Lesson Quiz

Multiplying and Dividing Rational Expressions

1. Simplify $\frac{x^2 - 2x}{(x-2)(x+3)}$.

   Ⓐ $\frac{1}{(x-2)(x+3)}$
   Ⓑ $\frac{2x}{x+3}$
   Ⓒ $\frac{x}{x+3}$
   Ⓓ $\frac{x-2}{x+3}$

2. Simplify $\frac{(x+5)(x^2-25)}{(x+5)^2(x-5)^2}$. What is the domain?

   Ⓐ $\frac{x+5}{x-5}$; all real numbers except –5 and 5
   Ⓑ $\frac{1}{x-5}$; all real numbers except –5 and 5
   Ⓒ $\frac{1}{x+5}$; all real numbers except 0
   Ⓓ $\frac{x-5}{x+5}$; all real numbers except –5

3. Simplify $\frac{ab^2 - a^2b}{ab} \cdot \frac{a^2b^2}{a-b}$.

   Ⓐ $-a^2b^2$; $a \neq 0$; $b \neq 0$
   Ⓑ $\frac{a^2b^2}{a-b}$; all real numbers except 0
   Ⓒ $\frac{b-a}{ab}$; $a \neq 0$ or b; $b \neq 0$ or a
   Ⓓ $ab^2 - a^2b$; $a \neq 0$ or b; $b \neq 0$ or a

4. What is the quotient of $\frac{9-x^2}{3x}$ and $\frac{x^2+6x+9}{3x}$?

   Ⓐ $\frac{x-3}{3-x}$
   Ⓑ $\frac{x-3}{x+3}$
   Ⓒ $\frac{3-x}{x+3}$
   Ⓓ $\frac{x+3}{x-3}$

5. The area of a rectangle is $\frac{x^2-4}{2x}$ in.^{2} and its length is $\frac{(x+2)^2}{2}$ in. What is the width in inches?

   Ⓐ $\frac{x(x+2)}{x-2}$
   Ⓑ $\frac{x+2}{x(x+2)}$
   Ⓒ $\frac{x-2}{x(x-2)}$
   Ⓓ $\frac{x-2}{x(x+2)}$

enVision® Algebra 2 • Assessment Sourcebook

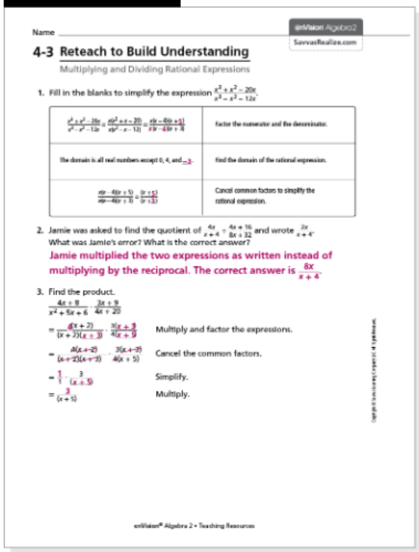
DIFFERENTIATED RESOURCES

I = Intervention   O = On-Level   A = Advanced

Reteach to Build Understanding I

Provides scaffolded reteaching for the key lesson concepts.

MathXL® for School


Name

4-3 Reteach to Build Understanding

Multiplying and Dividing Rational Expressions

1. Fill in the blanks to simplify the expression.

Factor the numerator and the denominator.

Find the domain of the rational expression.

Cancel common factors to simplify the rational expression.

2. Jamie was asked to find the quotient. What was Jamie's error? What is the correct answer?

Jamie multiplied the two expressions as written instead of multiplying by the reciprocal.

3. Find the product.

Multiply and factor the expressions.

Cancel the common factors.

Simplify.

Multiply.

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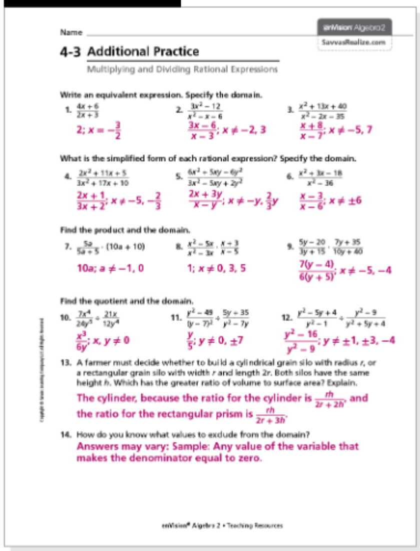
Additional Practice I O

Provides extra practice for each lesson.

MathXL® for School


Name

4-3 Additional Practice

Multiplying and Dividing Rational Expressions

Write an equivalent expression. Specify the domain.

What is the simplified form of each rational expression? Specify the domain.

Find the product and the domain.

Find the quotient and the domain.

13. A farmer must decide whether to build a cylindrical grain silo with radius r, or a rectangular grain silo with width r and length $2r$. Both silos have the same height h. Which has the greater ratio of volume to surface area? Explain.

The cylinder, because the ratio for the cylinder is $\frac{rh}{2r+2h}$ and the ratio for the rectangular prism is $\frac{rh}{2r+3h}$.

14. How do you know what values to exclude from the domain?

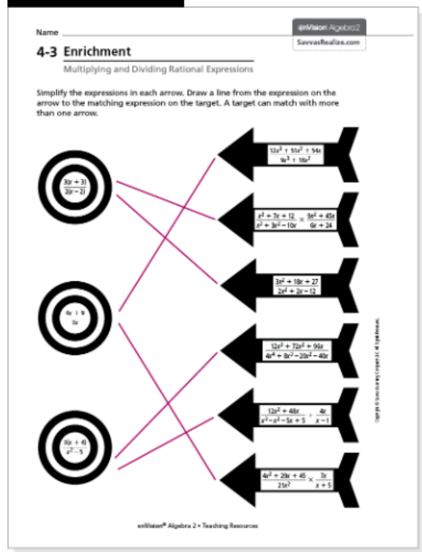
Answers may vary. Sample: Any value of the variable that makes the denominator equal to zero.

enVision® Algebra 2 • Teaching Resources

Enrichment O A

Presents engaging problems and activities that extend the lesson concepts.

MathXL® for School


Name

4-3 Enrichment

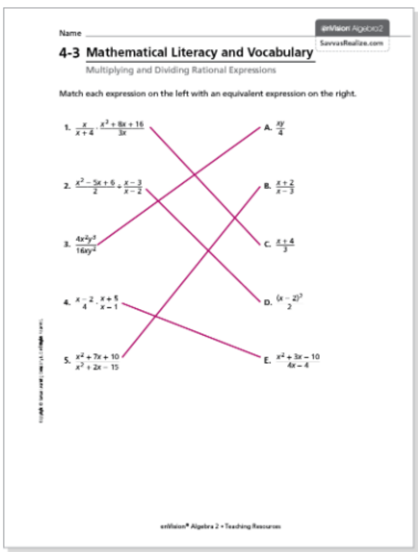
Multiplying and Dividing Rational Expressions

Simplify the expressions in each arrow. Draw a line from the expression on the arrow to the matching expression on the target. A target can match with more than one arrow.

enVision® Algebra 2 • Teaching Resources

Mathematical Literacy and Vocabulary I O

Helps students develop and reinforce understanding of key terms and concepts.


Name

4-3 Mathematical Literacy and Vocabulary

Multiplying and Dividing Rational Expressions

Match each expression on the left with an equivalent expression on the right.

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Digital Differentiated Resources

The **Reteach to Build Understanding**, **Additional Practice**, and **Enrichment** activities are available as digital assignments. These activities are automatically assigned when students complete the lesson quiz online and are automatically scored.